Bluestock Fintech: Mutual Fund Analytics Platform

Capstone Project Final Report — Data Engineering, ETL & Interactive Analytics

1. Executive Summary

The Indian mutual fund industry is experiencing rapid expansion, currently managing over ■81 lakh crore in AUM across more than 1,900 schemes. However, retail investors and financial analysts frequently face data fragmentation across asset management companies (AMCs), making objective risk-adjusted comparisons challenging. This capstone project delivers a comprehensive, end-to-end analytics platform. By ingesting raw data from AMFI, mfapi.in, and market indices, our system cleans, transforms, and loads normalized data into an optimized SQLite database (`bluestock_mf.db`), culminating in interactive Power BI and Streamlit dashboards for real-time portfolio insights.

2. Problem Statement & Objectives

- **Data Silos & Fragmentation:** Historical NAVs, AMC-level AUMs, SIP flows, and scheme details existed in isolated CSV and text formats. We unified them into a robust 5-table Star Schema.
- **Lack of Standardized Risk Metrics:** Investors lacked standardized tools to compute true risk-adjusted returns. We implemented automated calculation of 1/3/5-year CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown.
- **Benchmark Tracking Deficiencies:** We mapped scheme NAVs against NIFTY 50 and NIFTY 100 daily closing values to accurately compute tracking errors and market correlation.
- **Static Reporting Modernization:** Replaced legacy static reports with dynamic, self-serve interactive dashboards featuring multi-dimensional slicers and drill-downs.

3. Data Sources & ETL Architecture

Our automated data pipeline adheres to strict Data Engineering best practices:

1. **Extract (Ingestion):** Python scripts (`data_ingestion.py`, `live_nav_fetch.py`) extract raw data from AMFI daily feeds, mfapi.in historical REST APIs, and NSE/BSE index datasets into `data/raw/`.
2. **Transform (Cleaning):** Handled missing weekend and holiday NAV values using forward-filling (`ffill()`) across complete date reindexing. Standardized schema units to strictly distinguish between scheme-level AUM (in crores) and industry-wide AUM (in lakh crores).
3. **Load (Star Schema):** Loaded cleaned datasets into a normalized SQLite database (`data/db/bluestock_mf.db`) comprising `dim_fund`, `dim_date`, `fact_nav`, `fact_performance`, and `fact_transactions`.
4. **Analyze (Metrics):** Calculated advanced metrics including Historical VaR (95%), Conditional VaR (CVaR), and Sector Herfindahl-Hirschman Index (HHI) for concentration risk.

4. Key EDA & Performance Findings

- **Record SIP Inflows:** Monthly SIP contributions hit historical highs, crossing ■31,000+ crore, driven heavily by retail participation in mid and small-cap categories.
- **Risk-Return Tradeoff (Scorecard):** Our objective 0-100 Fund Scorecard revealed that while small-cap schemes delivered high raw returns, large-cap bluechip schemes offered superior Sortino ratios and significantly lower Maximum Drawdown during market volatility.
- **Sector Concentration (HHI):** Several thematic equity funds exhibited HHI scores exceeding 2,000, indicating severe over-concentration in Financial Services and IT sectors.
- **Cohort Dynamics:** The 2024 investor cohort represents the largest total AUM contribution, whereas the 2025 cohort displays higher transaction frequency with lower average ticket sizes.

5. Dashboard Implementation & Visual Analytics

The visual presentation layer spans two robust platforms: an interactive Power BI dashboard (`dashboard/bluestock_mf.pbix`) and a Python-native Streamlit web application (`dashboard/app.py`). Every dashboard page features at least two functional slicers (Category, Risk Grade, and AMC selection), enabling stakeholders to drill into fund performance, analyze rolling 90-day Sharpe trends, and assess SIP continuity risk scores.

6. Project Limitations & Actionable Recommendations

- **Limitations:** Historical NAV feeds do not reflect intra-day price volatility or real-time expense ratio revisions. Additionally, investor transaction logs are synthetically modeled from aggregate geographic distributions (`T30` vs `B30` cities).
- **Recommendation 1 (Pipeline Automation):** Schedule the master `etl_pipeline.py` script as a daily automated cron job or via Apache Airflow to continuously refresh NAV histories.
- **Recommendation 2 (Portfolio Optimization):** Upgrade the standalone fund recommender to incorporate Markowitz Efficient Frontier modeling, allowing investors to build multi-fund portfolios along optimal risk-return boundaries.